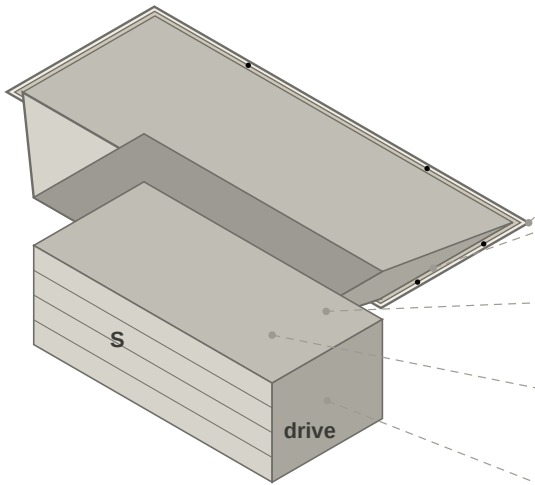
Sample Elementary RTU Duct Transition Schedule

Sample ISD, Sample Elementary School · Example Mechanical · field sketches by Pat Doe

Fittings (one per page following):

RTU-1	Supply · 60×24 to 40×20 · 18" tall · sep flex 40×20
RTU-1	Return · 60×18 to 40×20 · 18" tall · sep flex 40×20
Room 101	Supply · 30×16 to 32×14 · 40" tall · sep flex 32×14
Room 101	Return · 30×16 to 36×12 · 40" tall · sep flex 36×12
Room 102	Supply · 18×14 to 18×10 · sep flex 18×10 · sep sq-to-round 10"ø
Room 102	Return · 20×14 to 16×10 · sep flex 16×10 · sep sq-to-round 10"ø
Room 103	Transition 30×16 to 18×18, 32", right plumb (Qty 2) · sep flex 18×18

Every drop: 1" flange (4 sides) at curb top, S-cleat + drive on the girth joints, and 1.5" internal duct liner. Each flex is a separate piece sized to its drop bottom; square-to-round is separate. Inches. Field verify.



1" flange (4 sides)
screw to curb

Top opening
60 × 24

1.5" duct liner
internal

Drop bottom
40 × 20

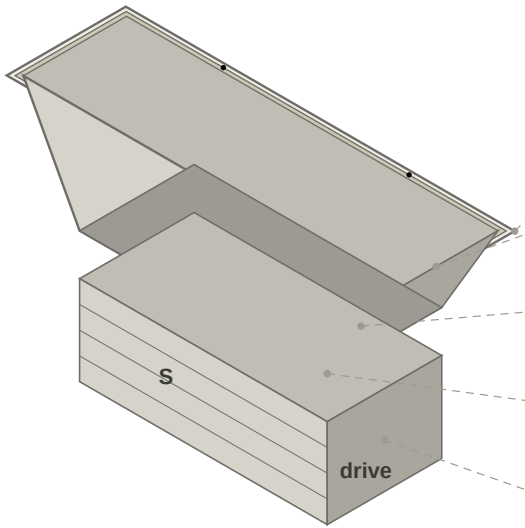
Flex (separate)
40 × 20

RTU-1 · Supply · transition + separate flex · Qty 1

Top (curb) 60 × 24 Bottom 40 × 20 Drop 18" tall · 1.5" liner

Flush left, taper right; depth 24 to 20 centered

Joints: S-cleat + drive (1" flange at curb top only)



1" flange (4 sides)
screw to curb

Top opening
60 × 18

1.5" duct liner
internal

Drop bottom
40 × 20

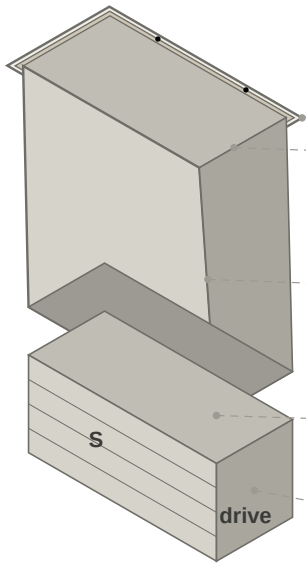
Flex (separate)
40 × 20

RTU-1 · Return · transition + separate flex · Qty 1

Top (curb) 60 × 18 Bottom 40 × 20 Drop 18" tall · 1.5" liner

Concentric (centered)

Joints: S-cleat + drive (1" flange at curb top only)



1" flange (4 sides)
screw to curb

Top opening
30 × 16

1.5" duct liner
internal

Drop bottom
32 × 14

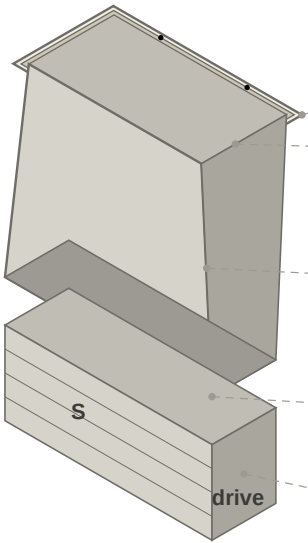
Flex (separate)
32 × 14

Room 101 · Supply · transition + separate flex · Qty 1

Top (curb) 30 × 16 Bottom 32 × 14 Drop 40" tall · 1.5" liner

Left wall plumb; depth 16 to 14 centered

Joints: S-cleat + drive (1" flange at curb top only)



1" flange (4 sides)
screw to curb

Top opening
30 × 16

1.5" duct liner
internal

Drop bottom
36 × 12

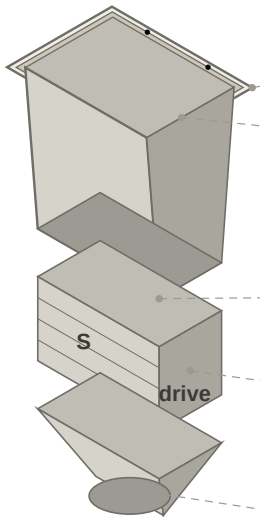
Flex (separate)
36 × 12

Room 101 · Return · transition + separate flex · Qty 1

Top (curb) 30 × 16 Bottom 36 × 12 Drop 40" tall · 1.5" liner

Right wall plumb; left flares 6" left

Joints: S-cleat + drive (1" flange at curb top only)



1" flange (4 sides)
screw to curb

Top opening
18 × 14

Drop bottom
18 × 10

Flex (separate)
18 × 10

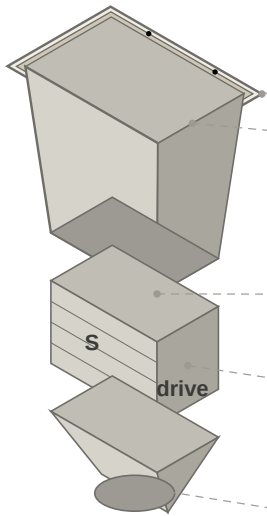
Square-to-round
separate, to 10"ø

Room 102 · Supply · drop + flex + square-to-round (3 separate pieces) · Qty 1

Top (curb) 18 × 14 Drop to 18 × 10, 24" tall · 1.5" liner

Separate flex 18 × 10 between drop and square-to-round (all separate)

Square-to-round to 10"ø, ht ~10" assumed (confirm) · S + drive joints



1" flange (4 sides)
screw to curb

Top opening
20 × 14

Drop bottom
16 × 10

Flex (separate)
16 × 10

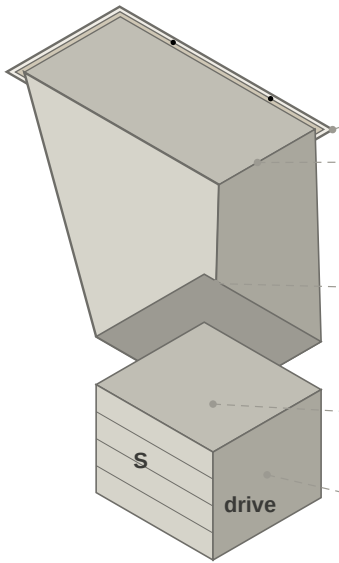
Square-to-round
separate, to 10"ø

Room 102 · Return · drop + flex + square-to-round (3 separate pieces) · Qty 1

Top (curb) 20 × 14 Drop to 16 × 10, 24" tall · 1.5" liner

Separate flex 16 × 10 between drop and square-to-round (all separate)

Square-to-round to 10"ø, ht ~10" assumed (confirm) · S + drive joints



1" flange (4 sides)
screw to curb

Top opening
30 × 16

1.5" duct liner
internal

Drop bottom
18 × 18

Flex (separate)
18 × 18

Room 103 · transition side + separate flex · Qty 2

Top (curb) 30 × 16 Bottom 18 × 18 Drop 32" tall · 1.5" liner

Right wall plumb; left tapers 12"

Joints: S-cleat + drive (1" flange at curb top only)